Physics Notes from Action Principles

Srikrishnaa Vadivel

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Chapter 1

Generating Functionals

A unified view starts with a configuration ϕ and an action $S[\phi]$. The quantum generating functional is

$$Z[J] = \int \mathcal{D}\phi \, \exp\left(\frac{i}{\hbar}S[\phi] + \frac{i}{\hbar}\int J\phi \, d^d x\right).$$

Classical equations follow from stationary phase: $\delta S/\delta\phi = 0$. Correlators follow by source differentiation:

$$\langle \phi(x_1)\cdots\phi(x_n)\rangle = \frac{1}{Z[0]}\left(\frac{\hbar}{i}\right)^n \frac{\delta^n Z}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}.$$

Chapter 2

Classical Mechanics and Statistical Mechanics

For generalized coordinates $q^i(t)$,

$$S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t)dt, \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} = 0.$$

Imaginary time $t = -i\tau$ turns oscillatory weights into Boltzmann weights:

$$Z(\beta) = \mathrm{Tr}\, e^{-\beta H} = \int_{\phi(\beta\hbar)=\phi(0)} \mathcal{D}\phi\, e^{-S_E[\phi]/\hbar}.$$

This connects quantum amplitudes, finite-temperature field theory, and statistical thermodynamics.

Chapter 3

Electromagnetism and Relativity

Electromagnetism uses A_μ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$:

$$S_{\text{EM}} = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu \right).$$

Variation gives $\partial_\mu F^{\mu\nu} = J^\nu$ plus the Bianchi identity $\partial_{[\lambda} F_{\mu\nu]} = 0$. Special relativity is encoded by the Minkowski interval $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$.

Chapter 4

General Relativity

The Einstein–Hilbert action is

$$S_{\mathrm{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g}(R - 2\Lambda) + S_m[g, \psi].$$

Metric variation gives

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

The geometry is determined by the Levi-Civita connection and curvature tensor, while matter enters through $T_{\mu\nu} = -(2/\sqrt{-g})\delta S_m / \delta g^{\mu\nu}$.

Chapter 5

Quantum Field Theory

QED has

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \quad D_\mu = \partial_\mu + ieA_\mu.$$

QCD replaces $U(1)$ by $SU(3)$:

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + \bar{q}(i\gamma^\mu D_\mu - m)q.$$

The Standard Model is based on $SU(3)_C \times SU(2)_L \times U(1)_Y$ with a Higgs field whose vacuum expectation value breaks electroweak symmetry.

Chapter 6

Strings and Effective Theories

The Polyakov action for a string embedding $X^\mu(\sigma, \tau)$ is

$$S_P = -\frac{T}{2} \int d^2\sigma \sqrt{-h}\, h^{ab} \partial_a X^\mu \partial_b X_\mu.$$

At low energies, effective field theories organize all symmetry-compatible operators by dimension, keeping predictive control through a cutoff scale Λ.